

Seth Baird

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EDUCATION

- **Purdue University** West Lafayette, Indiana
Anticipated Graduation May 2027
Major in Computer Engineering Technology Minor in Information Technology
- **Memorial High School** Frisco, Texas
Graduated May 2023
Vocational Education: CS3: Data Structures and Algorithms, Mobile Application Development

INVOLVEMENT

- **Purdue Electric Racing - STM32 Software Team**
 - Created multiple HALs (Hardware abstraction layers) to simplify the codebase and make development easier
 - Reworked the CAN communication backend to use two busses simultaneously for new high-throughput motors
 - Utilized testing equipment such as CANable and the oscilloscope's built in signal decoding to test new hardware

SKILLS AND PROJECTS

- **Hardware Engineering**
 - Implemented a channel scan algorithm into my own superheterodyne FM radio using DSP and statistics
 - Fabricated a fully custom guitar amplifier with a CAD-drawn case and a purpose-built PCB laid out in KiCad
 - Built industrial control systems hands-on in lab and programmed PLCs using IEC 61131-3 ladder logic in FTDW
 - Designed and simulated logic circuits and HDL code for Altera FPGAs in Quartus
 - Leveraged Solidworks, OpenSCAD, and a Raspberry Pi Pico running Rust to create a replica arcade game controller
- **Software Engineering and Information Technology**
 - Developed an android app that integrates with a car's engine computer to provide data and insights to the driver
 - Deployed a bash script using cron to automate bidirectional encrypted data transfer for off-site backups over VPN
 - Implemented a domain filtering DNS server to block advertisements and malicious sites on all network devices
 - Created over 20 proof-of-concept iOS and Android apps to gain familiarity with Swift and Kotlin

WORK EXPERIENCE

- **SMS InfoComm (Wistron Corporation)** Dallas, TX
NVIDIA Server Repair Technician May 2024 - August 2024
 - **System Repair and Troubleshooting** Diagnosed, refurbished, repaired, and built NVIDIA HGX 8-GPU units.
 - **Hardware Validation and Analysis** Used proprietary diagnostic tools to identify errors and discrepancies within motherboard subsystems such as power delivery, signal integrity, and FPGAs before executing a repair.
 - **Team Workflow Optimization** Developed a Python script to automate thousands of lines of by-hand data entry into the company's antiquated database, saving my team 80 man-hours per week
- **Jawa.gg** Remote
Verified Seller October 2021 - August 2023
 - **Top 10 Seller in Overall Performance** Recognized by CEO Amanda Stefan for driving Jawa's growth.
 - **Computer Sales and Commissions** Built and sold dozens of tailored desktop, laptop, and Mac computers to end users at different price points and configurations.
 - **Part Refurbishment and Sales** Repaired, tested, and refurbished parts and sold them to PC builders and end users through Jawa's marketplace.